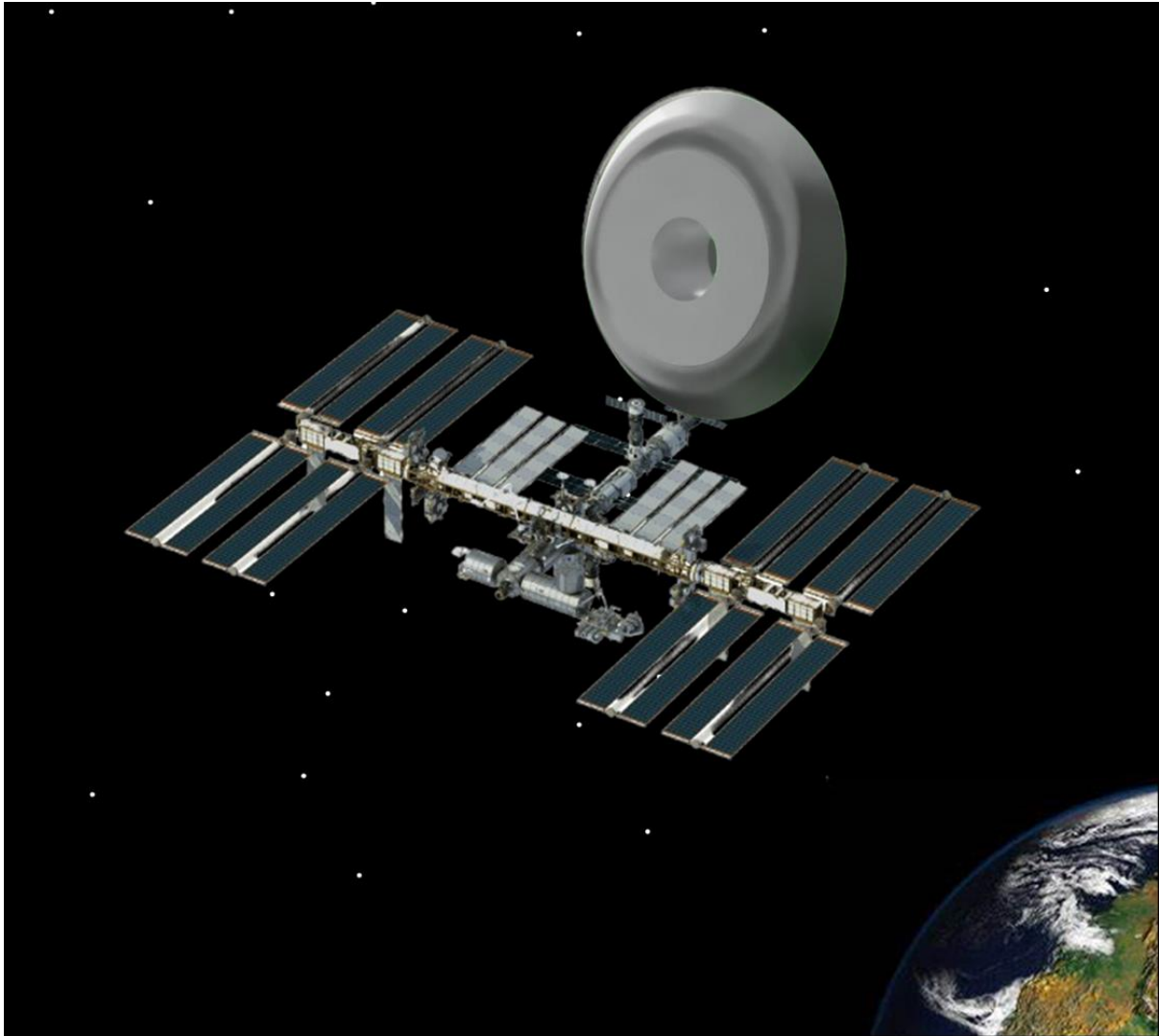




### Contents

1. Summary
2. General Concept
3. Habitat example layout
4. Benefits
5. Outline Plan

### Appendix

- Basic calculations
- Alternative print options
- Initial prototype

### 1. Summary

Having the ability to construct large rotating space facilities relatively simply and cheaply and with a scalable methodology will be a huge bonus towards a sustainable off world habitation.

The primary objectives are:

- Before the end of its useful life, use the ISS as a platform to construct a roughly 40m diameter enclosed rotating habitat in one piece using an additive manufacturing method.
- Demonstrate the potential of this manufacturing method for constructing much larger structures.

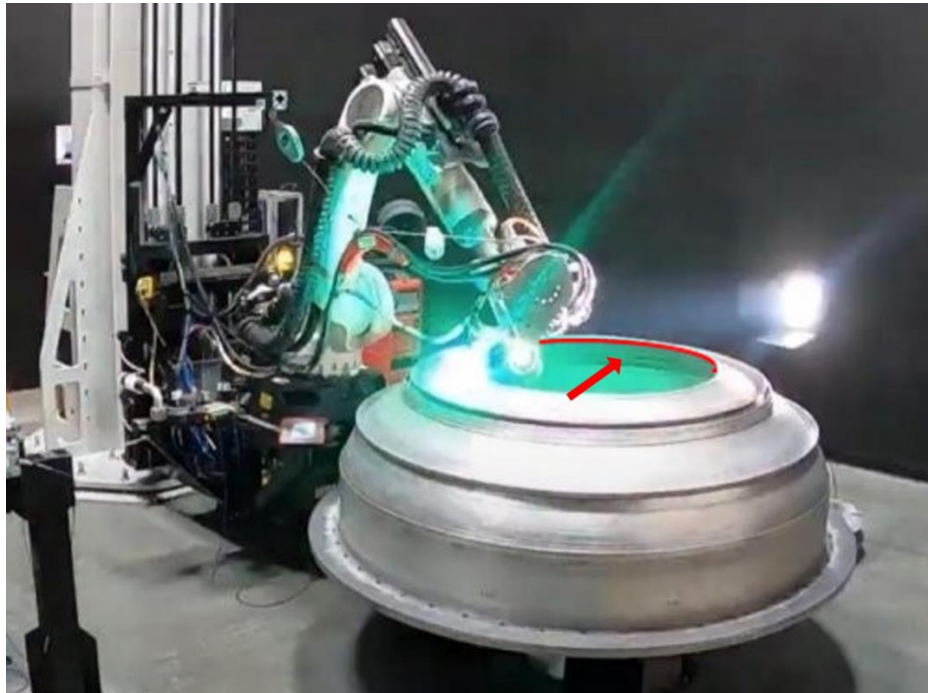
Milestones to achieving this being

1. Terrestrial design and testing of 3D print methodology
2. ISS testing of 3D print methodology

### 2. General concept

The additive manufacturing technique is very similar to that used today for various space applications and combines constant rotational movement of the object being manufactured with small movements of the print head after each revolution.

It is entirely possible to extend and growth a larger structure than the original base.



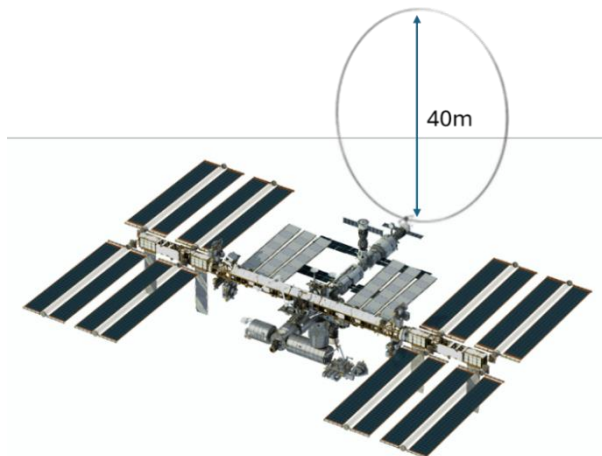
Imagine the surface indicated in red being the starting point and outside diameter of the habitat being constructed.

A narrow ring (pre-manufactured in sections) is assembled (without eva) and rotated and used as the initial print base upon which the habitat “grows” from the outside inwards.

## ISS based construction of rotating habitat

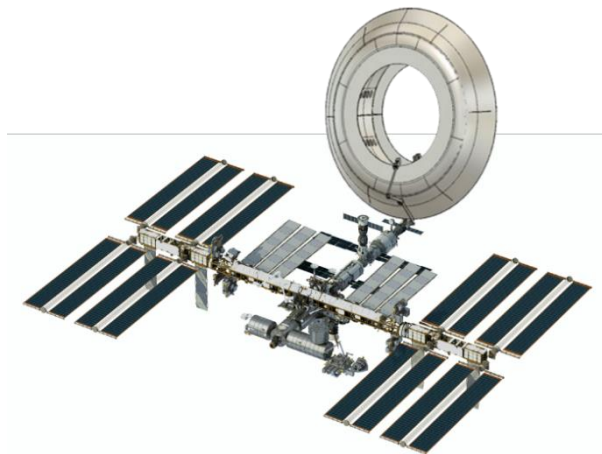
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### Start (Earliest 2028)



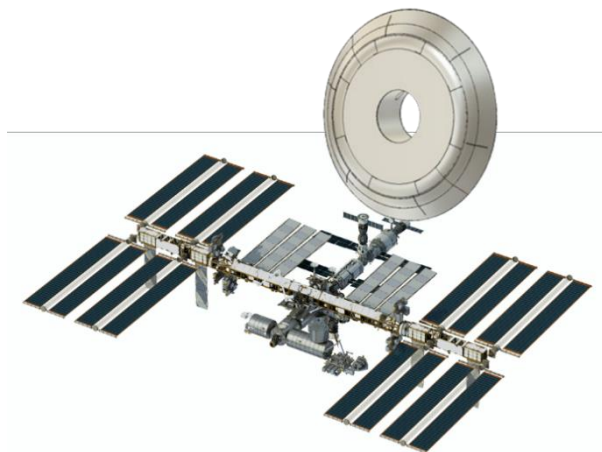
Seed ring assembled prior to print arms being assembled. This will form the surface upon which the habitat will be constructed.

### Progress (possibly 1-2 years)



Part way through construction, arms on either side of the ring position print heads. Internal fitment can start before the structure is fully completed

### Complete (with central docking)

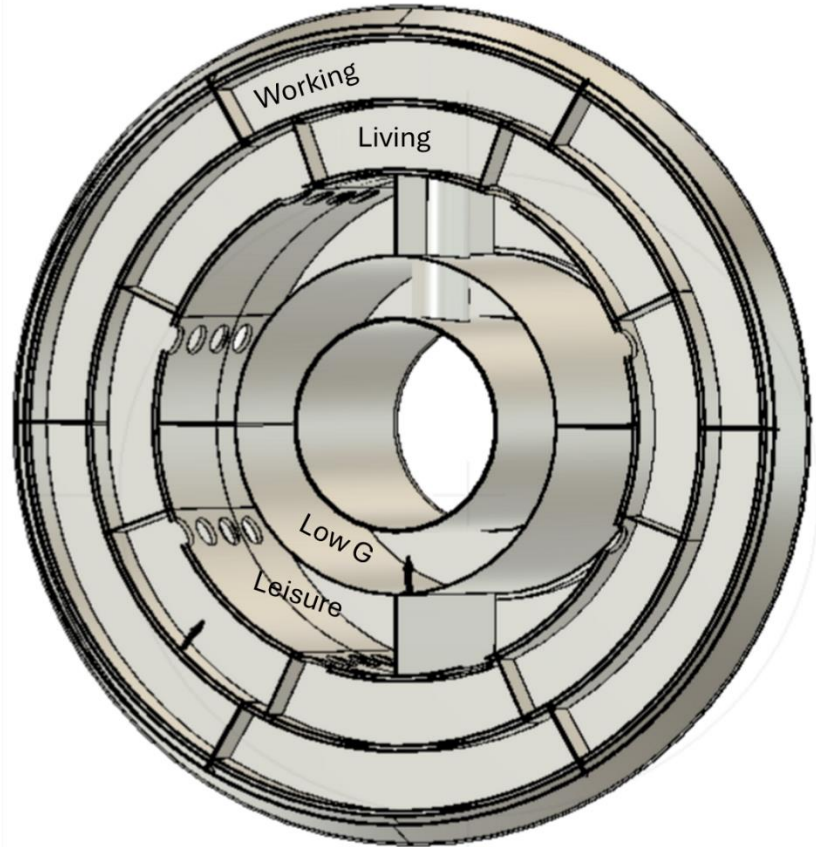
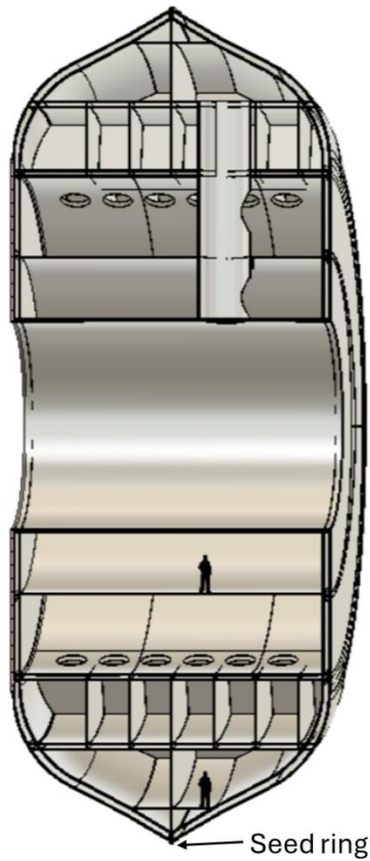


Completed structure ready for final fitment and release

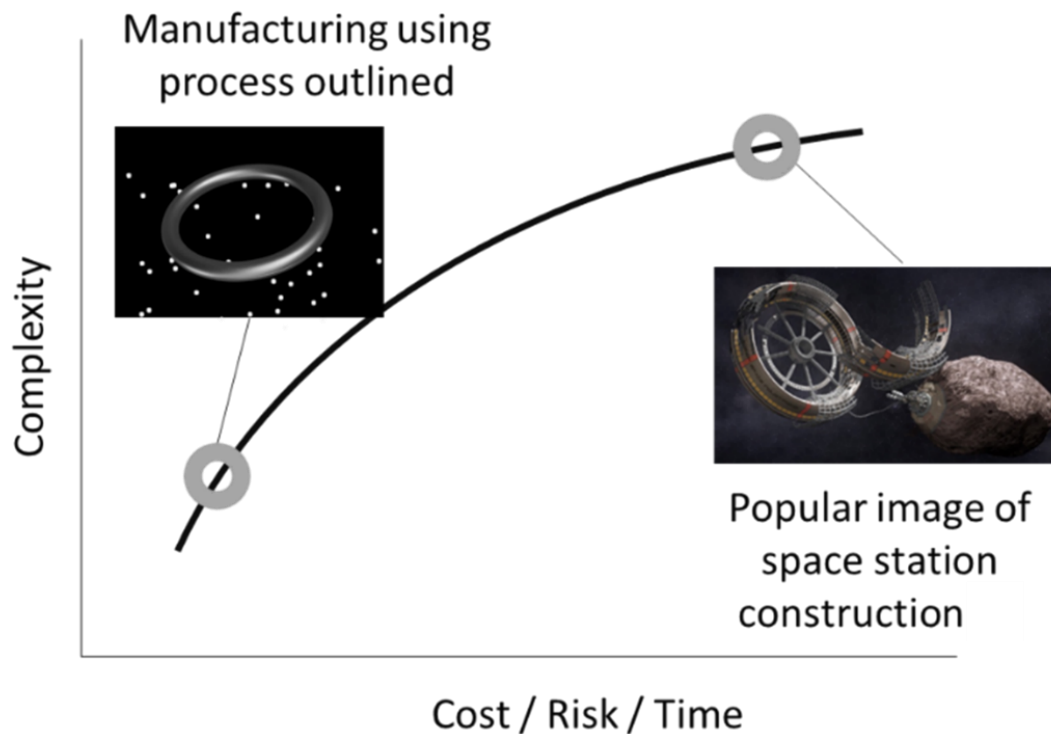
### 3. Habitat example layout

For illustration only, design has not started in this area.

Central hole could accommodate starship or similar.



### 4. Benefits



#### 4.1 Manufacturing Advantages:

- **Simplicity** - with only a small number of moving parts required in the manufacturing process the production facilities will be comparatively **cheap** (when considered against other imagined orbital construction methods) and have less failure points.
- Infrequent, small movements of the print head – less inertia to overcome, less wear – therefore an expectation of high mechanical reliability
- Other than materials feed, the main manufacturing process is **automated** (there will be a need to install bulkheads and all other internal fitting but the main structure is built in one piece automatically).
- The manufacturing process is highly **scalable** and capable of using different materials.
- Real time monitoring of print quality and ability to correct or compensate

#### 4.2 Habitat Advantages:

- Less parts and made “in house” being significantly cheaper
- The number of components and therefore interfaces reduced by an estimated 2-3 orders or magnitude (assuming somewhere between 1,000 and 10,000 individual components would be required for the equivalent structure using conventional methods).

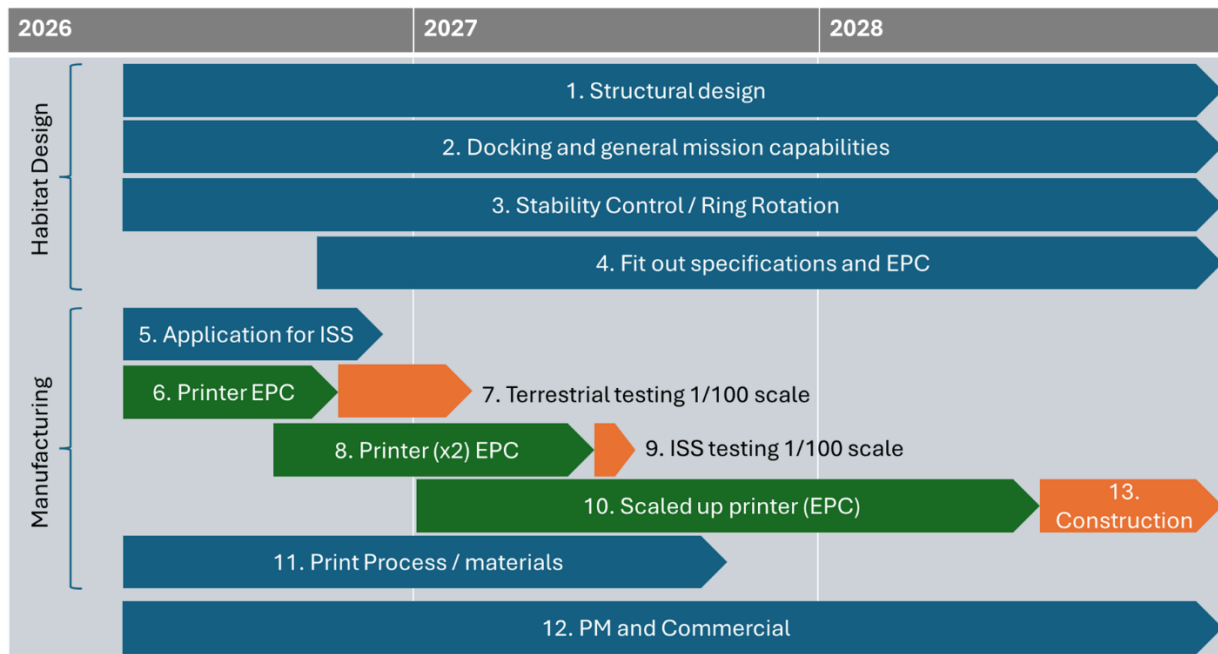
## ISS based construction of rotating habitat

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- **Very long life expectancy** (failure mode unclear)
- The structure has **fewer interfaces** to consider, making the design process comparatively **cheap** and **quick**
- The design can be done in parallel with manufacture thus making it **quicker overall**
- By building the internal services directly into the main infrastructure there is **lower failure risk** either through design or in service – simplicity is your friend

### 5. Outline Plan

The plan targets terrestrial testing of a 100<sup>th</sup> scale print facility commencing this year, ISS testing in 2026 and start of construction in 2027.



#### 5.1. Structural design

In relative terms, this is simple, literally one component and the design can be totally changed during the print process (obviously for the subsequent layers), although it is expected that multiple materials will be used to provide suitable containment surfaces for water and other fluids.

Potentially with the right insulator material power lines can be printed directly into the structure or in conduit format.

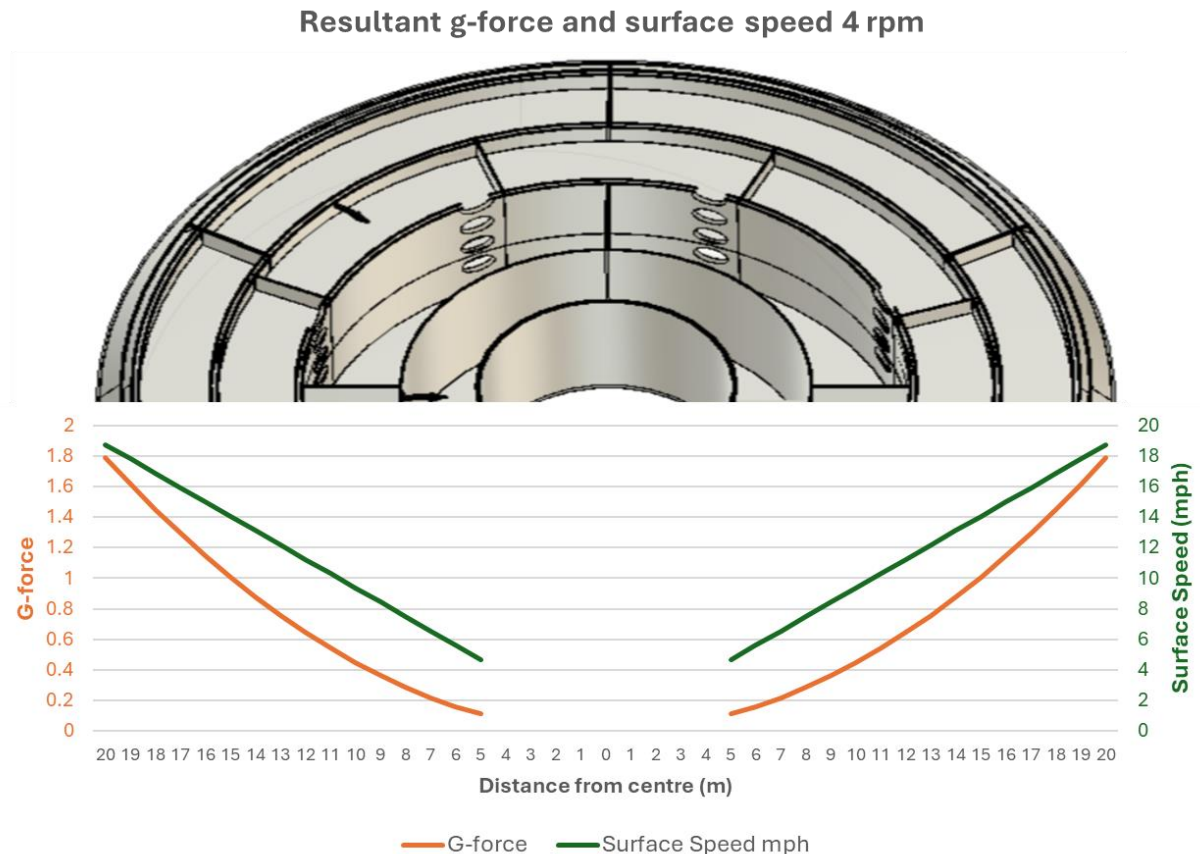
Bespoke slicer software will be required to be developed.

Significant modelling will be required to ensure the heat added to the model via the printing and solar radiation does not cause distortions or print issues.

#### 5.2. Docking and general mission capabilities

For a ring size of 40m and a mass of 635 tonnes (see appendix) a 12 tonne offset mass caused by a dragon capsule or similar might require counterbalancing (with water) and central docking is preferred at this point pending work to understand edge docking better including stability modelling.





The required rotational speed to maintain a roughly 1 g average at 4 rpm is significantly higher than would be used during the print process where print head speed is the limiting factor. High end printers are currently able to achieve 1 m/s.

The printing operation would run at about 0.2 rpm or one revolution every five minutes, generating negligible g forces and allowing safe access to the structure for fit out during construction.

### 5.3. Stability Control / Ring Rotation

Theoretical modelling is required to understand a number of questions

- Dynamics of rotating ring (is there any danger of an end swap?)
- What happens when it gets struck by micro meteorites?
- Implication of ISS movement during construction
- Method of maintaining rotation (thrusters – when are these fitted and where?)
- Water storage – ballast repositioning ring stability

### 5.4. Fit out specifications and EPC

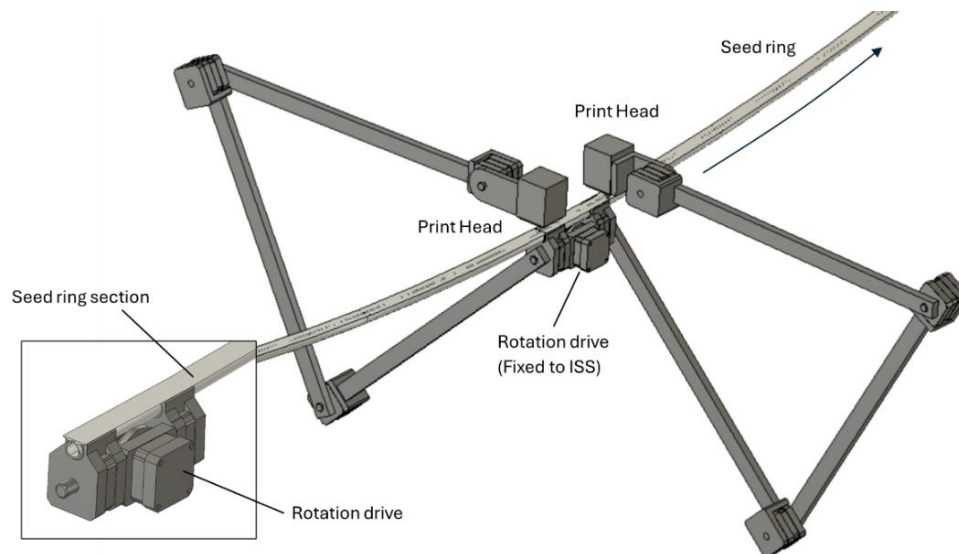
There is a good deal uncertainty regarding the final use and people onboard and responsibility for undertaking the fit-out activity.

A complete list is beyond the scope of this document but will include all life support, communications, power (the side of the habitat could be used to generate some energy but it is smaller than the array currently on the ISS and additional power will be required either through a nuclear battery or potentially beamed using microwaves from an adjacent array), food storage, food preparation, bathrooms, bedrooms, working spaces, exercise and leisure, docking, hatches, furnishing, lighting, computing and more.

### 5.5. Application for ISS time

It is unclear exactly the timescales required for this activity and approvals, the plan assumes minimal delays and availability of facilities.

### 5.6. Printer EPC (Engineering, procurement and construction)



At the most basic level the printer consists of a guide to keep the seed ring orientated correctly, a drive motor to rotate the ring and articulated arms to position the print heads (likely to require multiple to achieve reasonable print times).

The position of printing is always exactly in the same place relative to the pivot point for the arms and there are no large movements – print quality therefore expected to be high.

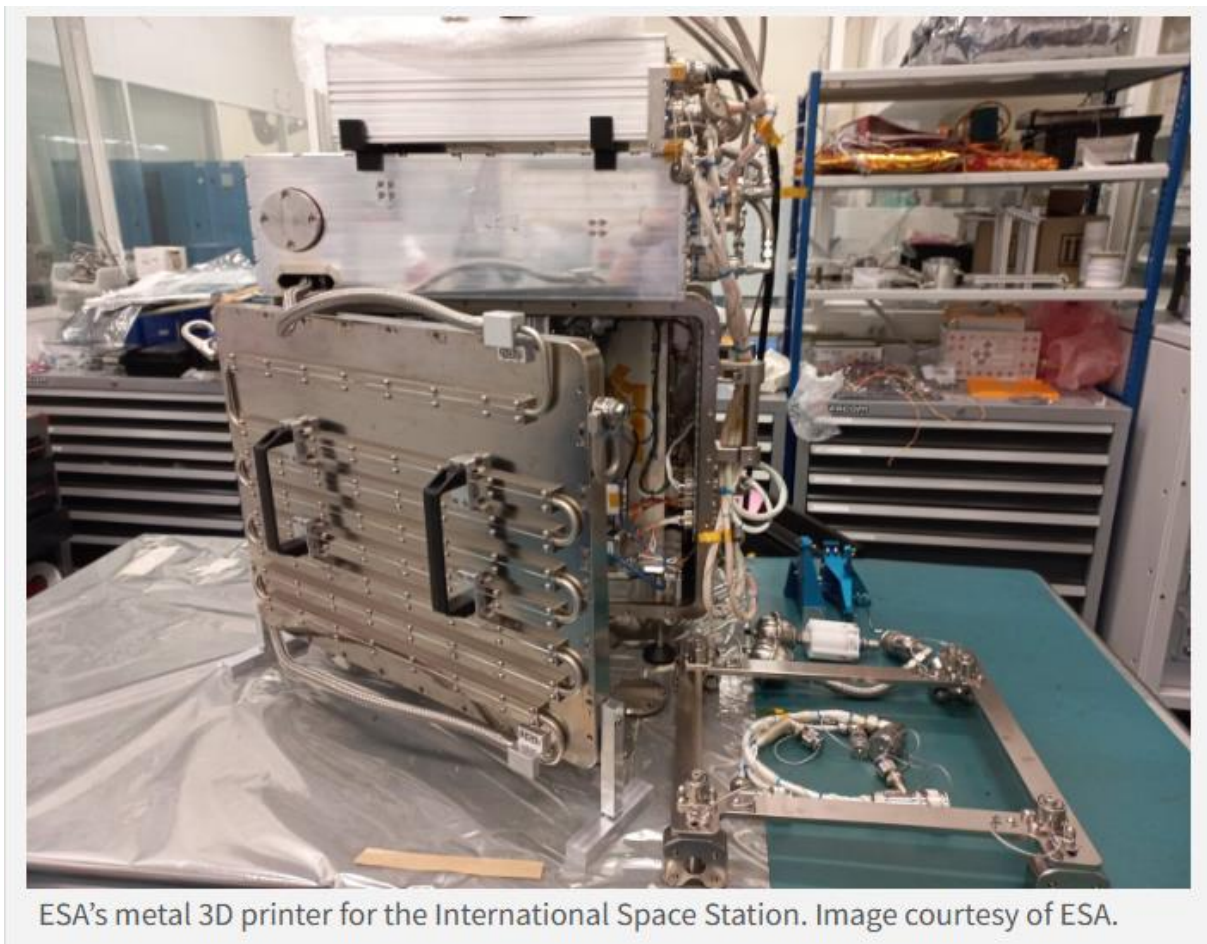
There are multiple commercially available robotic arms that are capable of performing the basic movements required and the print deposition process is relatively mature and tested on board the ISS.

## ISS based construction of rotating habitat

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Readily available robotic arm



ESA's metal 3D printer for the International Space Station. Image courtesy of ESA.

### 5.7. Terrestrial testing 1/100 scale

Terrestrial testing will be undertaken to prove the concept and evaluate the slicing software

### 5.8. ISS Printer EPC

A version will be developed for use on the ISS that will be capable of printing a 400 mm diameter version in a micro gravity and vacuum environment and that can be tested for stability.

### 5.9. ISS testing 1/100 scale

Testing to include:

- Print reliability
- Thermal camera recording during printing
- Quality of material (to be determined on earth)
- Free float
- Testing stability – adding masses
- Testing stability – simulate impacts with larger objects (same relative inertia)
- Forces on mount point

### 5.10. Scaled up printer (EPC)

Additional requirements including:

- Emergency release capabilities
- Filament feed and heating
- Lidar surface scans in real time

### 5.11. Print Process / materials

Building from existing literature and experience to determine and best material and source.

### 5.12. PM and Commercial

Depending on the exact commercial structure there are a range of stages and states of construction in which the structure can be handed to the tenant (assuming a lease investment repayment model).

This could range from literally the bare single piece structure through to ready for fitment of customer equipment to ready to occupy.

### 5.13. Construction

Construction time will be dependant on the design and the number and size of print heads, print depth and speed.

Clearly with a print depth in mm it is not going to be possible to generate good mating surfaces and it is expected that appropriate filling and bonding techniques

## **ISS based construction of rotating habitat**

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would be used where interfaces are required to connect to the internal facilities. All internal surfaces can be accurately scanned during manufacture to allow mating parts to be manufactured.

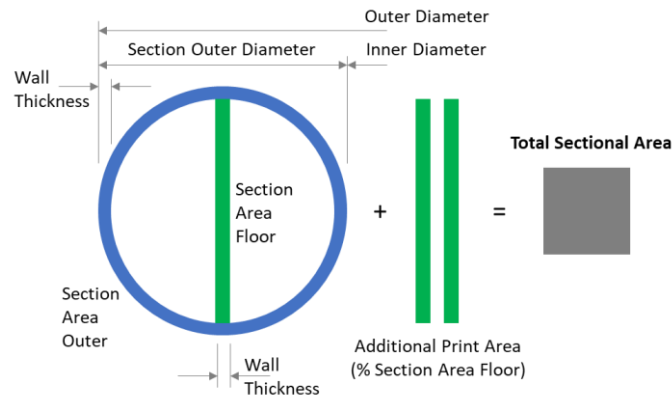
## Appendix

### Basic calculations (see link to generic model page 21)

Table 1 shows three similar options with values in the unshaded cells. Calculated or referenced values are indicated in the shaded cells.

A ring of this size has 14 times the internal volume of ISS and with 40 people of board would offer 335m<sup>3</sup> of space per person compared to 155 in the ISS (both these numbers exclude space requirements for facilities, equipment etc. and should not be considered personal living space). A double skin 20mm wall thickness is used.

Print volume is calculated by subtracting the torus volume generated by the internal dimensions from that generated by the external dimensions and then adding additional volume for internal features and print supports, assumed as a percentage of the volume occupied by the central floor (a cylinder). Print volume is used to calculate mass and flow rates. Total sectional area is used later in to calculate print time.



Approximate calculations for total sectional area or swept area

| Dimensions             | Units | Option 1 | Option 2 | Option 3 | ISS | Notes                                                                                   |
|------------------------|-------|----------|----------|----------|-----|-----------------------------------------------------------------------------------------|
| Outer Diameter         | m     | 40       | 40       | 40       | 73  | Inside diameter of the Seed Ring. ISS dimension relates to maximum length               |
| Section Outer Diameter | m     | 15       | 15       | 15       |     | Outside diameter of the section - for the purposes of simplicity assumed to be circular |
| Inner Diameter         | m     | 10       | 10       | 10       |     |                                                                                         |
| Wall Thickness         | m     | 0.04     | 0.04     | 0.04     |     |                                                                                         |
| Additional Print Area  | %     | 330%     | 330%     | 330%     |     | For further inner floors / walls as a percentage of                                     |



## ISS based construction of rotating habitat

|                                         |    |        |        |        |     |                                                                  |
|-----------------------------------------|----|--------|--------|--------|-----|------------------------------------------------------------------|
|                                         |    |        |        |        |     | the central floor section area                                   |
| Internal Volume                         | m3 | 13,410 | 13,410 | 13,410 | 932 |                                                                  |
| Total Print Volume                      | m3 | 469    | 469    | 469    |     |                                                                  |
| People on Board                         |    | 40     | 40     | 40     | 6   | Possibly constrained by bed space                                |
| Average volume per person               | m3 | 335    | 335    | 335    | 155 |                                                                  |
| Floor Space per person at 2.5 m ceiling | m2 | 134    | 134    | 134    |     | Approximate calculation does not consider curvature of structure |
| Personal Space                          | m2 | 31     | 31     | 31     |     | From model                                                       |

**Table 1.** Main dimension calculated for a ring of 40m in diameter. Three versions have different external wall thicknesses from 20 – 100 mm.

Ring material and associated mass and launch factors are shown in Table 2. Options 1 and 2 consider an Aluminium structure, Option 3 considers one made from High Density ABS (as a proxy for a plastic structure of material tbd).

| Material Requirements         | Units     | Option 2  | Option 2  | Option 3         | ISS | Notes                     |
|-------------------------------|-----------|-----------|-----------|------------------|-----|---------------------------|
| Main Structure Material       |           | Al (3003) | Al (3003) | High Density ABS |     |                           |
| Material Density              | kg/m3     | 2,705     | 2,705     | 1,100            |     | From Table                |
| Wall Design Density           | %         | 50%       | 50%       | 50%              |     |                           |
| Total Mass                    | tons      | 635       | 635       | 258              | 420 |                           |
| Raw Materials Production Rate | tons /day | 0.4       | 1.1       | 0.3              |     |                           |
| Falcon Heavy Lifts            |           | 10        | 10        | 4                |     | 64 Ton lift capacity      |
| Launch Cost                   | \$m       | 150       | 150       | 150              |     | Assumed \$150m per launch |
| Lifts frequency               | days      | 151.3     | 60.5      | 186.0            |     |                           |
| Lift Cost                     | \$bn      | 1.5       | 1.5       | 0.6              |     |                           |

**Table 2.** Materials selected and associated impact on total mass and shipping costs.

Print time is calculated by combining the geometric properties of the habitat, specifically the swept area with the size of the nozzle and the print speed as shown below.

## ISS based construction of rotating habitat

$$\begin{array}{l}
 \text{Total Sectional Area} \quad \text{Print Area} \\
 \text{[Grey Square]} \quad / \quad \text{[Diagram: Print Depth x Nozzle Diameter]} \times \text{Number Nozzles} = \text{Number Revolutions} \\
 \\
 \text{Print Head Speed} / \text{Ring Circumference} = \text{Rotational Speed} \\
 \\
 \text{Number Revolutions} / \text{Rotational Speed} = \text{Time}
 \end{array}$$

Calculations of total time. Total Sectional Area = Swept Area

Materials production rate and lift frequency based upon the manufacturing characteristics shown in table 3 is well within near term or current capabilities. Average wall density impacts the mass calculations and launches but not the total print time as it is assumed that the swept area is the same as for a solid.

| Manufacturing Process          | Units              | Option 1 | Option 2 | Option 3 | ISS | Notes                                                                                                                                                               |
|--------------------------------|--------------------|----------|----------|----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Print Speed                    | m/s                | 0.4      | 0.4      | 0.4      |     | How fast the printer head moves relative to the surface being printed<br>Current home 3D print speed. Aluminium extrusion in range of 5 - 50 m/min (0.08 - 0.8 m/s) |
| Nozzle Diameter                | mm                 | 4        | 5        | 4        |     |                                                                                                                                                                     |
| Print Depth                    | mm                 | 2        | 2        | 2        |     |                                                                                                                                                                     |
| Nozzle volumetric flow rate    | cm <sup>3</sup> /s | 3        | 4        | 3        |     | For all Nozzles                                                                                                                                                     |
| Rotational Speed               | RPM                | 0.19     | 0.19     | 0.19     |     | Calculated from the print head speed at the outer diameter                                                                                                          |
| G-force outer edge (RCF)       |                    | 0.0008   | 0.0008   | 0.0008   |     | Induced by rotation of ring                                                                                                                                         |
| Number of Print Head Nozzles   |                    | 4        | 8        | 8        |     | Number of nozzles for the main construction material - separate print head are assumed for other materials                                                          |
| Flow Rate per Nozzles          | kg/s               | 0.009    | 0.011    | 0.004    |     |                                                                                                                                                                     |
| Flow Rate All Nozzles          | kg/s               | 0.035    | 0.087    | 0.028    |     | Used in power consumption calculations                                                                                                                              |
| Print Efficiency               | %                  | 90%      | 90%      | 90%      |     | Time Printing vs Time Not Printing NOT INCLUDING wall design density                                                                                                |
| Number of Complete Revolutions | million            | 0.37     | 0.15     | 0.19     |     | Based upon the total sectional area divided by the print area (nozzle diameter x print depth x number of nozzles)                                                   |



## ISS based construction of rotating habitat

|            |       |       |     |     |  |                                                             |
|------------|-------|-------|-----|-----|--|-------------------------------------------------------------|
| Print Time | days  | 1,501 | 600 | 750 |  | Calculated using number of revolutions and rotational speed |
| Print Time | years | 4.1   | 1.6 | 2.1 |  |                                                             |

**Table 3.** Calculation concerning the manufacturing process. Input variables are constant for each of the options

Table 3 covers the manufacturing process with four key input variables print head speed, nozzle diameter and print depth used to determine the rotational speed of the ring during printing and the approximate print time.

An Aluminium alloy is the favoured material although the data is shown for pure aluminium. ABS has been included to demonstrate the relative differences in power requirement shown in Table 4.

| Power Requirements             | Units  | Option 1 | Option 2 | Option 3 | ISS | Notes                                                                                                                                              |
|--------------------------------|--------|----------|----------|----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Melt Energy                    | GJ/ton | 2.07     | 2.07     | 0.36     |     |                                                                                                                                                    |
| Total Energy                   | GJ     | 1,314    | 1,314    | 93       |     |                                                                                                                                                    |
| Process Efficiency             | %      | 70%      | 70%      | 70%      |     | Inefficiencies in overall system such melt vessel heat losses and power transmission losses                                                        |
| Continuous Power               | kW     | 102      | 256      | 15       | 120 | Calculated by assuming all nozzles are concurrently printing at the flow rates<br>For additional comparison nuclear power station range up to 8 MW |
| Solar energy in orbit          | W/m2   | 1,365    | 1,365    | 1,365    |     | The location of the manufacturing site relative to the position of the sun, earth and moon will impact this number                                 |
| Panel Efficiency               | %      | 30%      | 30%      | 30%      |     |                                                                                                                                                    |
| Theoretical Minimum Panel Size | m2     | 250      | 625      | 35       |     | Assumes continuous line of site to the sun                                                                                                         |
| Square Sides                   | m      | 16       | 25       | 6        | 50  | Dimensions of square panel                                                                                                                         |
| Diameter                       | m      | 18       | 28       | 7        | 56  | Equivalent size of the circular panel                                                                                                              |

## ISS based construction of rotating habitat

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| Panel Size Using ISS panel size / power ratio | m2 | 3,047 | 7,618 | 432 | 2500 | Assumes a worst case. |
|-----------------------------------------------|----|-------|-------|-----|------|-----------------------|
| Square Sides                                  | m  | 55    | 87    | 21  |      |                       |
| Diameter                                      | m  | 62    | 99    | 23  |      |                       |

**Table 4.** Power Requirements estimated based upon flow rates and material melt properties and converted into solar panel sizes.

There is a considerable discrepancy between the size of the panels on the ISS and associated power output figure with that calculated (with an estimated panel efficiency of 30%) in Table 4. It is assumed that this is be due to time in direct sunlight and efficiency differences and not a flaw in the theoretical calculation.

Theoretically a habitat with 21,000 m<sup>3</sup> of internal space and a 20mm wall thickness could be constructed in 8 ½ years (plus fitting time) with an ISS sized power supply and modest lift costs.

# ISS based construction of rotating habitat

## Alternative print options

### Dimensions

|                                         | Units          | Option 1 | Option 2 | Option 3 | ISS | Notes                                                                                                                                          | Source                                                                                                                                                                              |
|-----------------------------------------|----------------|----------|----------|----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outer Diameter                          | m              | 40       | 40       | 40       | 73  | Inside diameter of the Seed Ring. ISS dimension relates to maximum length                                                                      | <a href="https://en.wikipedia.org/wiki/Electrical_system_of_the_International_Space_Station">https://en.wikipedia.org/wiki/Electrical_system_of_the_International_Space_Station</a> |
| Section Outer Diameter                  | m              | 15       | 15       | 15       |     | Outside diameter of the section - for the purposes of simplicity assumed to be circular                                                        |                                                                                                                                                                                     |
| Inner Diameter                          | m              | 10       | 10       | 10       |     |                                                                                                                                                |                                                                                                                                                                                     |
| Wall Thickness                          | m              | 0.04     | 0.04     | 0.04     |     |                                                                                                                                                |                                                                                                                                                                                     |
| Section Area Outer                      | m <sup>2</sup> | 5.0      | 5.0      | 5.0      |     | Area of the outer walls required to be fully covered by the print head                                                                         |                                                                                                                                                                                     |
| Section Area Floor                      | m <sup>2</sup> | 1.6      | 1.6      | 1.6      |     | Area of the central floor area required to be fully covered by the print head                                                                  |                                                                                                                                                                                     |
| Additional Print Area                   | %              | 330%     | 330%     | 330%     |     | For further inner floors / walls as a percentage of the central floor section area                                                             |                                                                                                                                                                                     |
| Total Sectional Area                    | m <sup>2</sup> | 11.9     | 11.9     | 11.9     |     | This number is used in conjunction with the print nozzle size and print depth to calculate how many revolutions are required to cover the area |                                                                                                                                                                                     |
| Outer Torus Volume                      | m <sup>3</sup> | 13,879   | 13,879   | 13,879   |     |                                                                                                                                                |                                                                                                                                                                                     |
| Inside Torus Outer Diameter             | m              | 39.92    | 39.92    | 39.92    |     |                                                                                                                                                |                                                                                                                                                                                     |
| Inside Torus Inner Diameter             | m              | 10.08    | 10.08    | 10.08    |     |                                                                                                                                                |                                                                                                                                                                                     |
| Inside Torus Volume                     | m <sup>3</sup> | 13,732   | 13,732   | 13,732   |     | Layer thickness                                                                                                                                |                                                                                                                                                                                     |
| Torus Shell Volume                      | m <sup>3</sup> | 148      | 148      | 148      |     |                                                                                                                                                |                                                                                                                                                                                     |
| Central floor Volume                    | m <sup>3</sup> | 75       | 75       | 75       |     | Volume associated with a ring shaped floor at the widest point with a thickness equal to the outer wall                                        |                                                                                                                                                                                     |
| Additional Print Volume                 | m <sup>3</sup> | 322      | 322      | 322      |     |                                                                                                                                                |                                                                                                                                                                                     |
| Internal Volume                         | m <sup>3</sup> | 13,410   | 13,410   | 13,410   | 932 |                                                                                                                                                | <a href="https://www.nasa.gov/feature/facts-and-figures">https://www.nasa.gov/feature/facts-and-figures</a>                                                                         |
| Total Print Volume                      | m <sup>3</sup> | 469      | 469      | 469      |     |                                                                                                                                                |                                                                                                                                                                                     |
| People on Board                         |                | 40       | 40       | 40       | 6   | Possibly constrained by bed space                                                                                                              |                                                                                                                                                                                     |
| Average volume per person               | m <sup>3</sup> | 335      | 335      | 335      | 155 |                                                                                                                                                |                                                                                                                                                                                     |
| Floor Space per person at 2.5 m ceiling | m <sup>2</sup> | 134      | 134      | 134      |     | Approximate calculation does not consider curvature of structure                                                                               |                                                                                                                                                                                     |
| Personal Space                          | m <sup>2</sup> | 31       | 31       | 31       |     | From model                                                                                                                                     |                                                                                                                                                                                     |

### Manufacturing Process

|                                |                    |        |        |        |  |                                                                                                                                                                     |  |
|--------------------------------|--------------------|--------|--------|--------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Print Speed                    | m/s                | 0.4    | 0.4    | 0.4    |  | How fast the printer head moves relative to the surface being printed<br>Current home 3D print speed. Aluminium extrusion in range of 5 - 50 m/min (0.08 - 0.8 m/s) |  |
| Nozzle Diameter                | mm                 | 4      | 5      | 4      |  |                                                                                                                                                                     |  |
| Print Depth                    | mm                 | 2      | 2      | 2      |  |                                                                                                                                                                     |  |
| Nozzle volumetric flow rate    | cm <sup>3</sup> /s | 3      | 4      | 3      |  | For all Nozzles                                                                                                                                                     |  |
| Rotational Speed               | RPM                | 0.19   | 0.19   | 0.19   |  | Calculated from the print head speed at the outer diameter                                                                                                          |  |
| G-force outer edge (RCF)       |                    | 0.0008 | 0.0008 | 0.0008 |  | Induced by rotation of ring                                                                                                                                         |  |
| Number of Print Head Nozzles   |                    | 4      | 8      | 8      |  | Number of nozzles for the main construction material - separate print head are assumed for other materials                                                          |  |
| Flow Rate per Nozzles          | kg/s               | 0.009  | 0.011  | 0.004  |  |                                                                                                                                                                     |  |
| Flow Rate All Nozzles          | kg/s               | 0.035  | 0.087  | 0.028  |  | Used in power consumption calculations                                                                                                                              |  |
| Print Efficiency               | %                  | 90%    | 90%    | 90%    |  | Time Printing vs Time Not Printing NOT INCLUDING wall design density                                                                                                |  |
| Number of Complete Revolutions | million            | 0.37   | 0.15   | 0.19   |  | Based upon the total sectional area divided by the print area (nozzle diameter x print depth x number of nozzles)                                                   |  |
| Print Time                     | days               | 1,501  | 600    | 750    |  | Calculated using number of revolutions and rotational speed                                                                                                         |  |
| Print Time                     | years              | 4.1    | 1.6    | 2.1    |  |                                                                                                                                                                     |  |

### Material Requirements

| Main Structure Material       |                   | Aluminium ▼ | Aluminium ▼ | High Impact ABS ▼ |     |                                                                                                                                                                                                                                     |
|-------------------------------|-------------------|-------------|-------------|-------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Material Density              | kg/m <sup>3</sup> | 2,705       | 2,705       | 1,100             |     | From Table                                                                                                                                                                                                                          |
| Wall Design Density           | %                 | 50%         | 50%         | 50%               | 420 |                                                                                                                                                                                                                                     |
| Total Mass                    | tons              | 635         | 635         | 258               |     | <a href="https://www.nasa.gov/feature/facts-and-figures">https://www.nasa.gov/feature/facts-and-figures</a>                                                                                                                         |
| Raw Materials Production Rate | tons /day         | 0.4         | 1.1         | 0.3               |     |                                                                                                                                                                                                                                     |
| Falcon Heavy Lifts            |                   | 10          | 10          | 4                 |     | <a href="https://arstechnica.com/science/2018/02/three-years-of-sls-development-could-buy-86-falcon-heavy-launches/">https://arstechnica.com/science/2018/02/three-years-of-sls-development-could-buy-86-falcon-heavy-launches/</a> |
| Launch Cost                   | \$m               | 150         | 150         | 150               |     | <a href="https://arstechnica.com/science/2018/02/three-years-of-sls-development-could-buy-86-falcon-heavy-launches/">https://arstechnica.com/science/2018/02/three-years-of-sls-development-could-buy-86-falcon-heavy-launches/</a> |
| Lifts frequency               | days              | 151.3       | 60.5        | 186.0             |     |                                                                                                                                                                                                                                     |
| Lift Cost                     | \$bn              | 1.5         | 1.5         | 0.6               |     |                                                                                                                                                                                                                                     |

### Power Requirements

|                                               |                  |       |       |       |      |                                                                                                                                                    |
|-----------------------------------------------|------------------|-------|-------|-------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Melt Energy                                   | GJ/ton           | 2.07  | 2.07  | 0.36  |      |                                                                                                                                                    |
| Total Energy                                  | GJ               | 1,314 | 1,314 | 93    |      |                                                                                                                                                    |
| Process Efficiency                            | %                | 70%   | 70%   | 70%   |      | Inefficiencies in overall system such as melt vessel heat losses and power transmission losses                                                     |
| Continuous Power                              | kW               | 102   | 256   | 15    | 120  | Calculated by assuming all nozzles are concurrently printing at the flow rates<br>For additional comparison nuclear power station range up to 8 MW |
| Solar energy in orbit                         | W/m <sup>2</sup> | 1,365 | 1,365 | 1,365 |      | The location of the manufacturing site relative to the position of the sun, earth and moon will impact this number                                 |
| Panel Efficiency                              | %                | 30%   | 30%   | 30%   |      | <a href="https://www.solarenspace.com/solarens-space-solar/ssp-overview/">https://www.solarenspace.com/solarens-space-solar/ssp-overview/</a>      |
| Theoretical Minimum Panel Size                | m <sup>2</sup>   | 250   | 625   | 35    |      | Assumes continuous line of site to the sun                                                                                                         |
| Square Sides                                  | m                | 16    | 25    | 6     | 50   | Dimensions of square panel                                                                                                                         |
| Diameter                                      | m                | 18    | 28    | 7     | 56   | Equivalent size of the circular panel                                                                                                              |
| Panel Size Using ISS panel size / power ratio | m <sup>2</sup>   | 3,047 | 7,618 | 432   | 2500 | Assumes a worst case.                                                                                                                              |
| Square Sides                                  | m                | 55    | 87    | 21    |      |                                                                                                                                                    |
| Diameter                                      | m                | 62    | 99    | 23    |      |                                                                                                                                                    |

## ISS based construction of rotating habitat

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### Initial prototype



### Excel Model (generic annulus):

[https://drive.google.com/file/d/1huvJvDT9SeNQf\\_jpeFSetPIE2F3dQzUF/view?usp=sharing](https://drive.google.com/file/d/1huvJvDT9SeNQf_jpeFSetPIE2F3dQzUF/view?usp=sharing)

### Paper:

[https://www.researchgate.net/publication/346966266\\_Potential\\_route\\_for\\_the\\_construction\\_of\\_a\\_super\\_massive\\_space\\_habitat](https://www.researchgate.net/publication/346966266_Potential_route_for_the_construction_of_a_super_massive_space_habitat)

### Video:

<https://youtu.be/CyqPvjUyTnw>